



EC2101-1 – COMMUNICATION SYSTEMS – I
(E&C)

Max. Marks: 100

12. The scheme of modulation that consumes minimum bandwidth is
 A) DSB-SC B) DSB-FC
 C) SSB-SC D) FM
13. The figure of merit for the amplitude modulation scheme when depth of modulation is 30% is
 A) 0.3 B) 0.43
 C) 0.043 D) 0.111
14. A 400 W carrier is modulated to a depth of 75 percent. The total power for the SSB modulated signal is
 A) 512.5 W B) 112.5 W
 C) 200 W D) 56.25 W
15. The figure of merit for the DSB-SC modulation is
 A) 0.5 B) 1
 C) -1 D) infinity
16. The number of quantization levels for a PCM scheme is 64. A synchronizing pulse is added at the end of each code word. The number of bits used in the code word is
 A) 4 B) 5
 C) 6 D) 7
17. Let W be the bandwidth of message signal $m(t)$. AM signal would be recovered successfully if,
 A) $f_c > W$ B) $f_c > 2W$
 C) $f_c > 3W$ D) $f_c > 4W$
18. The signal is bandlimited to 3.6 kHz. The Nyquist's rate for the signal is
 A) 6 kHz B) 7.2 kHz
 C) 3.6 kHz D) 4 kHz
19. The amount of quantization error in the PCM scheme is
 A) 2Δ volts B) Plus Δ volts
 C) Minus Δ volts D) Plus or minus $\Delta/2$
20. Slope overload distortion will occur in delta modulation only if
 A) Step size is too small B) Step size is too large
 C) Signal voltage is very less D) Signal voltage is very high

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the generation of Amplitude Modulated wave using Square law modulator with relevant mathematical equations.	6	L*2	1	1
	b) A carrier wave with amplitude 12 V and frequency 10 MHz is amplitude modulated to 50 % level with a modulating signal of frequency 1 kHz. Determine: (i) the time domain equation for the amplitude modulated wave. (ii) the power required to transmit the modulated wave. (iii) the side band frequencies.	4	L3	1	1
	c) With neat circuit diagram and waveforms explain the envelope detector used to demodulate the message signal from an amplitude modulated signal.	6	L2	1	1
2.	a) A double side band suppressed carrier modulated signal is applied to the coherent detector which is supplied with a locally generated carrier signal $\cos(2\pi f_c t + \phi)$. (i) Draw the block diagram of coherent detector (ii) Derive the expression for the signal at the output of the coherent detector. (iii) Discuss the effect of ϕ on the demodulated signal.	6	L3	1	1
	b) The Figure Q2(b) shows the frequency multiplication and heterodyne section of an FM transmitter. Determine the carrier frequency and frequency deviation at points 1,2 and 3. Assume that at point 3, additive frequency is selected by the mixer.				

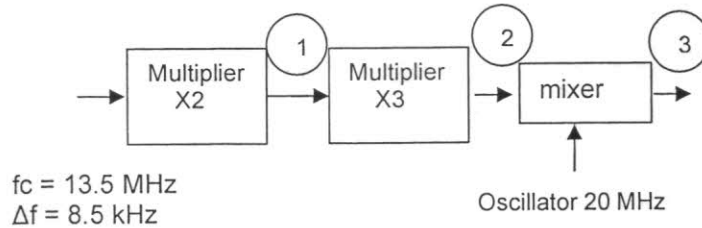


Figure Q2(b)

6 L2 2 1

- c) An amplitude modulated waveform has the form:

$$x(t) = 10[1 + 0.5 \cos 200\pi t + \cos 400\pi t] \cos 2000\pi t.$$

Sketch the amplitude spectra of $x(t)$.

4 L3 1 1

3. a) An angle modulated signal is defined by:

$$s(t) = 10 \cos[2\pi \times 10^6 t + 5 \sin 2000\pi t + 10 \sin 3000\pi t] \text{ volts.}$$

Determine the following:

- The instantaneous frequency
- The frequency deviation
- The deviation ratio
- the approximate transmission bandwidth
- The power contained in the modulated signal

10 L3 2 1

- b) Explain the generation of double sideband suppressed carrier (DSB-SC) signal using Ring modulator.

6 L2 1 1

Unit – II

4. a) Two $30 \text{ k}\Omega$ and a $50 \text{ k}\Omega$ resistors are connected in series. For an effective noise bandwidth of 100 kHz and temperature of 290 K , determine the rms value of the Thermal noise voltage.
- b) Three telephone links are connected in tandem connection. Two links have the SNR of 60 dB and the third one has SNR of 30 dB respectively. Determine the overall SNR for the cascade connection and comment on your result.
- c) Derive the expressions for power spectral density and auto correlation function for the white noise process passed through an ideal low pass filter.

4 L3 3 1

6 L3 3 1

6 L2 3 1

5. a) Discuss the noise in DSB receiver with a receiver model using coherent detection. Derive the figure of merit of the receiver.
- b) Explain about the pre-emphasis and de-emphasis techniques used in frequency modulation scheme with neat diagrams and equations.

10 L2 4 1

6 L2 4 1

6. a) Derive the expression for figure of merit of AM scheme. Discuss the role of message power P and amplitude sensitivity K_a .
- b) Derive the expression for the overall noise factor of cascaded two port networks.

10 L2 4 1

6 L2 3 1

Unit – III

7. a) A signal $g(t)$ which is band limited to $W \text{ Hz}$ is sampled at f_s . Derive the expression for the sampled version of the signal and also recover the original signal from its samples. Comment on the aliasing effect.
- b) A signal $g(t) = 10 \cos 20\pi t$. $\cos 200\pi t$ is ideally sampled at $f_s = 250 \text{ samples/sec}$.
- Sketch the spectrum of message signal and sampled signal
 - Specify the Nyquist rate for the signal $g(t)$
 - Specify the cut-off frequency of ideal reconstruction filter.

10 L2 5 1

6 L3 5 1

8. a) Explain Delta Modulation scheme with neat diagrams and discuss the types of errors observed in Delta Modulation scheme.

8 L2 5 1

- b) A PCM system uses a uniform quantizer followed by a 7 bit binary encoder. The bit rate of the system is equal to 50×10^6 bps.
- (i) Determine the maximum message bandwidth for which the system operates satisfactorily.
- (ii) Determine the output SNR when a full load sinusoidal modulating wave of frequency 1MHz is applied to the output.
- c) Assume a speech signal with a frequency of 3 kHz and maximum amplitude of 1 V. The speech signal is applied to delta modulator with its bit rate at 25kbps. Determine an appropriate step size.

4	L3	5	1
4	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
